CNC Procedure 2016_032	**Oct 10th** **2016**
Procedure for **<u>Dis-ASSEMBLY of</u>** **<u>α-i/α-iB Series Amplifier</u>**	[Applicable for all Models of α-iA/B Spindle Amplifiers]

Dis-ASSEMBLY of α-i/α-iB Series Amplifier

Note:

- ✓ Please practise proper safety measures during this activity
- ✓ High Power carrying devices are exposed on opening the drive enclosure
- ✓ Use proper tools to avoid damage to the PCB's, screw's.
- ✓ **Please take care for DC CHARGE LED indication near to DC bus bar**

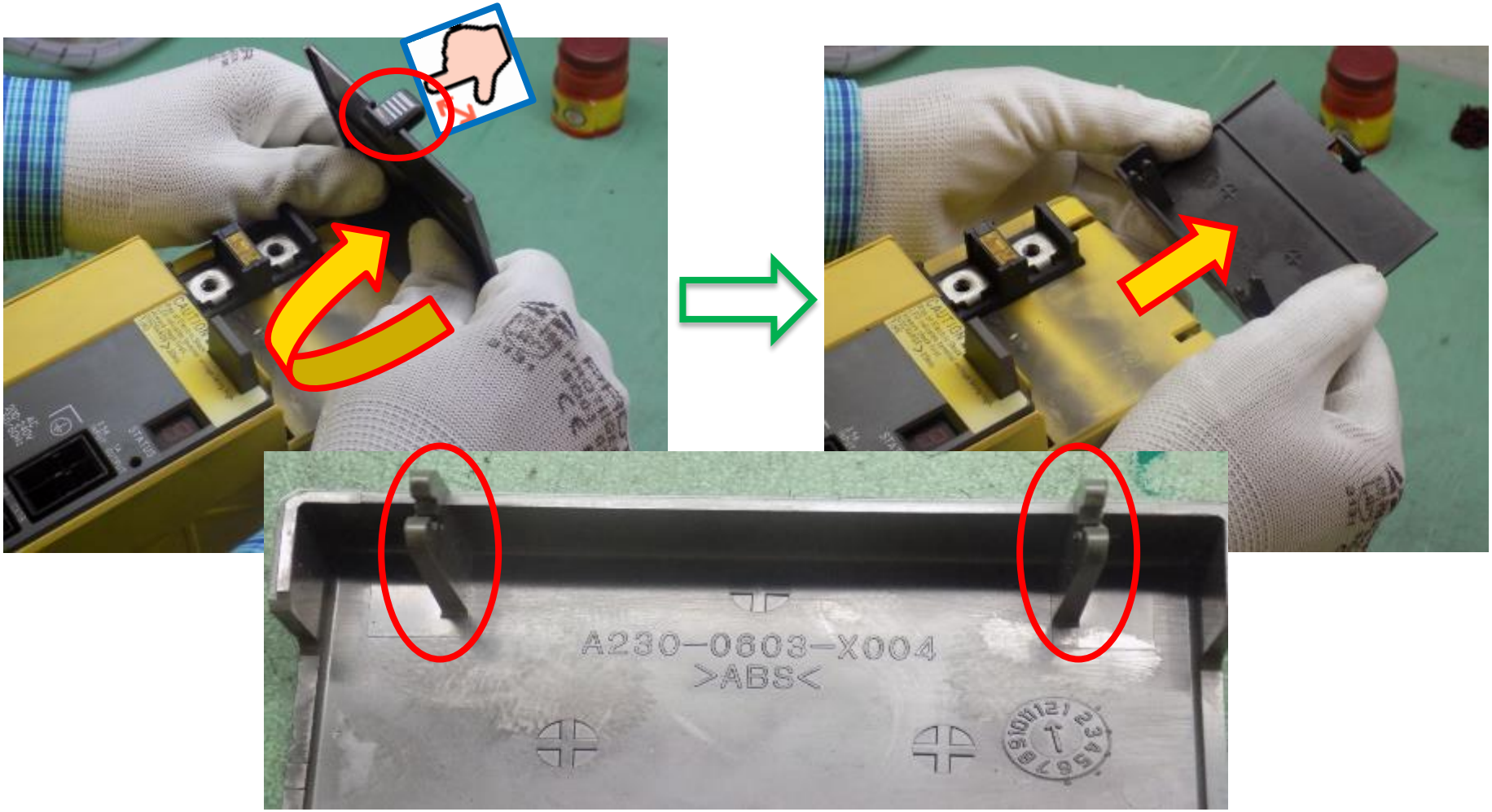
Applicable Amplifiers

For all *α-i/α-iB* Series Amplifier

α-i	*α-iB*
A06B-6110-Hxxx	A06B-6200-Hxxx
A06B-6111-Hxxx	A06B-6202-Hxxx
A06B-6112-Hxxx	A06B-6220-Hxxx
A06B-6114-Hxxx	A06B-6222-Hxxx
A06B-6115-Hxxx	A06B-6230-Hxxx
A06B-6116-Hxxx	A06B-6240-Hxxx
A06B-6117-Hxxx	A06B-6250-Hxxx
A06B-6120-Hxxx	A06B-6252-Hxxx
A06B-6121-Hxxx	A06B-6270-Hxxx
A06B-6122-Hxxx	A06B-6272-Hxxx
A06B-6124-Hxxx	A06B-6280-Hxxx
	A06B-6282-Hxxx
	A06B-6290-Hxxx

Check for the Amplifier specification on the DC link flip at top portion of the unit.

Removal of DC Link Flip (αi Series)



Lift the DC Link flip cover by pressing the highlighted hinge and the pull the flip upward

Ensure that the flip cover lock is not damaged

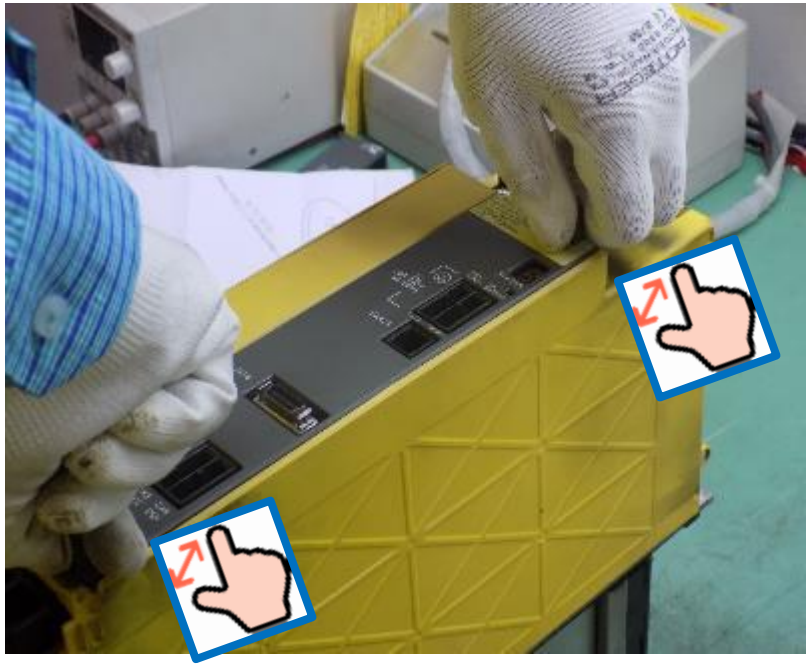
Removal of DC Link Flip (αi-B Series)

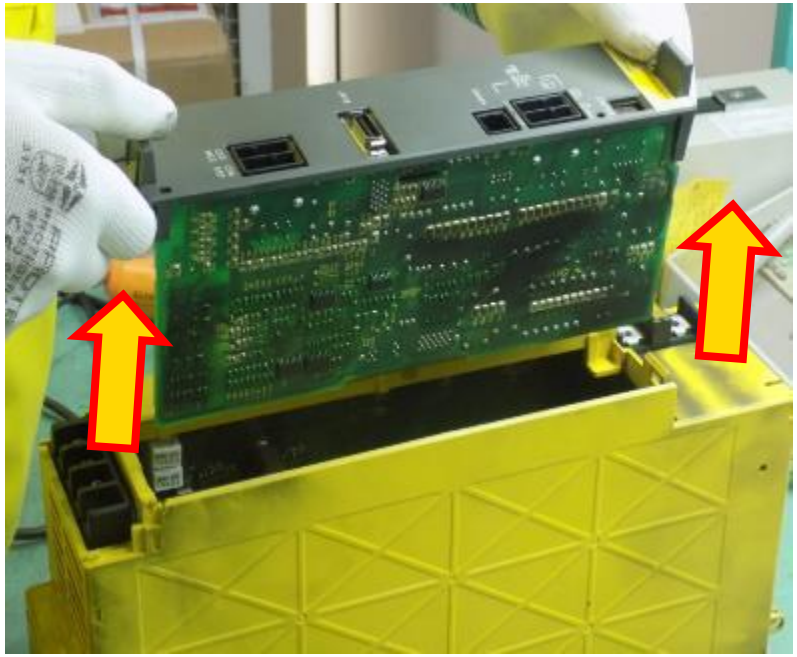
Please locate screw driver to small round window on the DC link cover flip
Turn the screw driver on CCW to unlock the flip from front cover
Then lift the DC Link flip cover by pressing the highlighted hinge and the pull the flip upward
Ensure that the flip cover lock is not damaged

Removal of Terminal Flip

- After this next comes the removal of the terminal cover flip
- Apply a little on the side edge of the flip and pull it upward direction and thus removing the flip safely

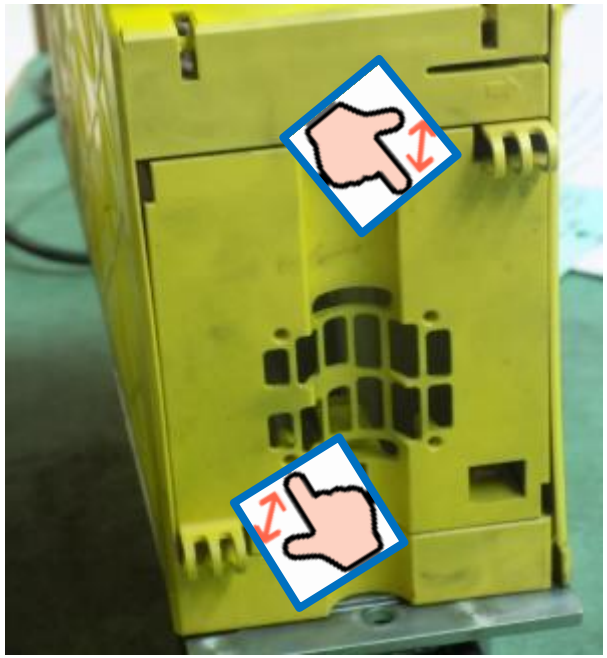
Removal of Face plate Cover

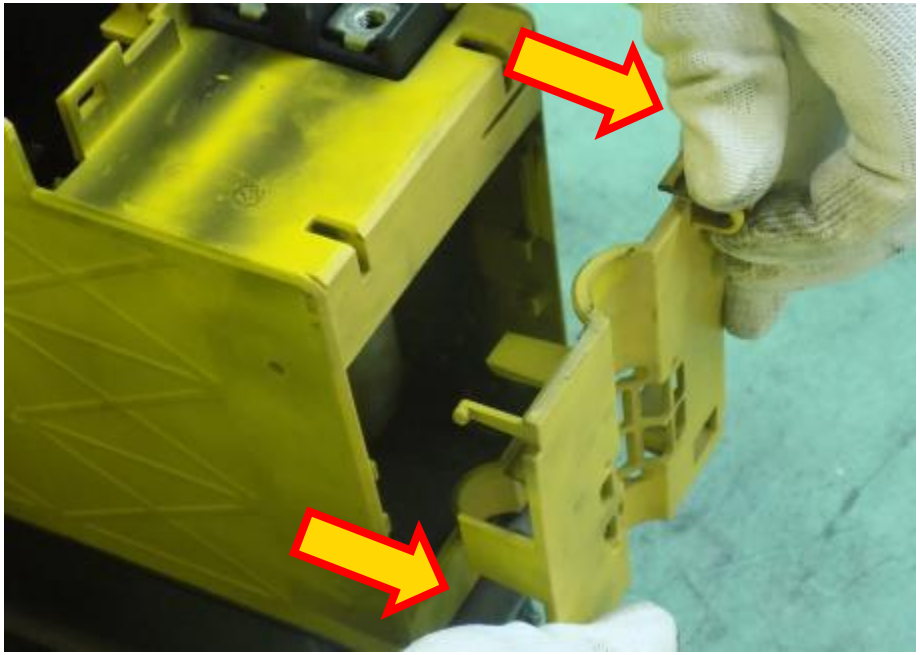




- Then comes the removal of the face plate from the drive
- Gently press the side edge flips of the face plate
- Applying little pressure to it and lift it from the location
- The face plate is lifted up and the control card comes along with the face plate when pulled upward from the case

Removal of Inverter/Internal Cooling Fan Cover





- The next process is to remove the invertor cooling fan cover from the top portion of the amplifier
- Press the side edges of the case and gently pull it up from the case
- Ensure to fix in same orientation during drive assembly

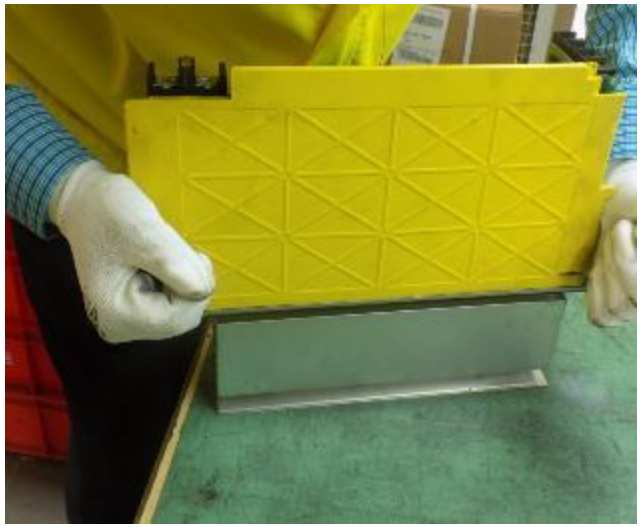
Removal of Radiator Fan

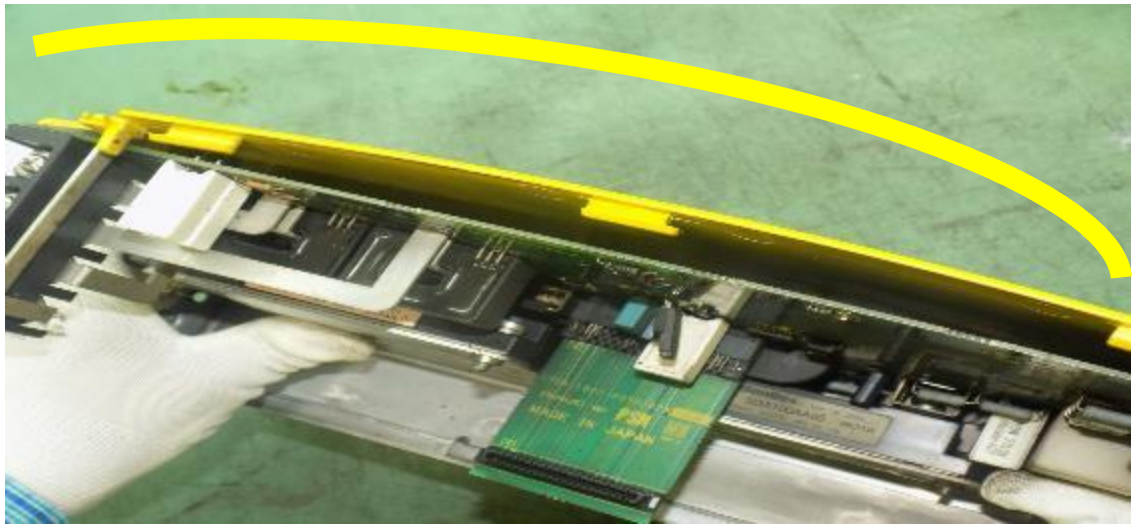
- The next stage is removal of the radiator cooling fan base from radiator
- Remove the screws from both left and right side of the fan mounting plate base using a star screw driver.
- Then gently pull the fan base from location ensure picture.
- This fan to be removed safely as it should not damage the connector pins of the fan nor pin holder in the power.

Removal of Plastic Cover LH Side

- To remove the case from the drive, pull the encircled locks.
- After unlocking the side locks of the drive, gently tap on the upper surface of the case. This will allow the side cover to open up.

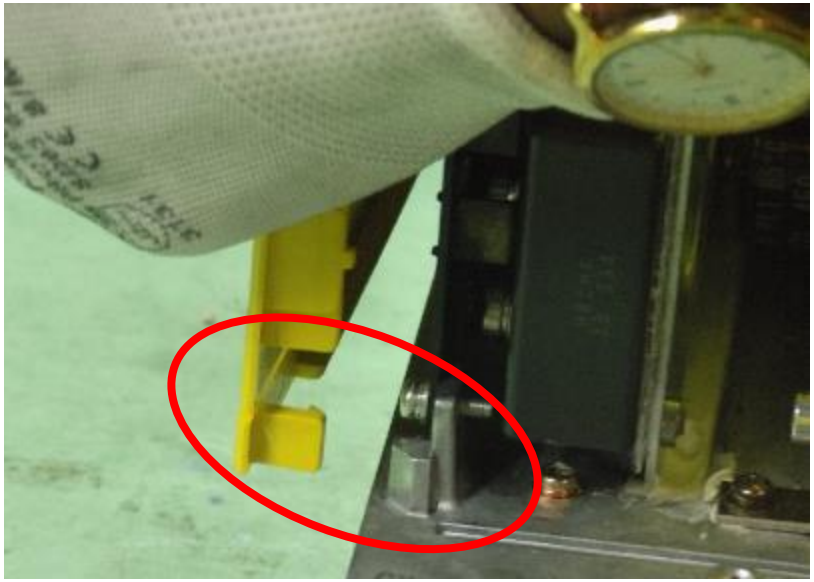
Removal of Plastic Cover RH Side





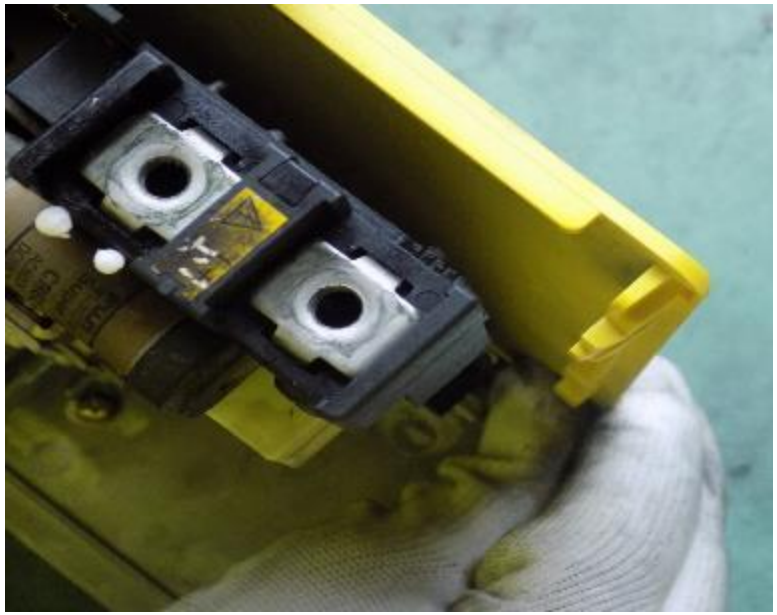
- The next step is to remove the side flip of the case
- It is the most sensitive part among drive covers which has locking at middle and on both sides, if not handled carefully may lead to the damage of the lock or cover may breaking apart
- Press the covers from both sides as shown in figure
- Know the cover bends as shown in top view at this condition lock in the center part of cover is released

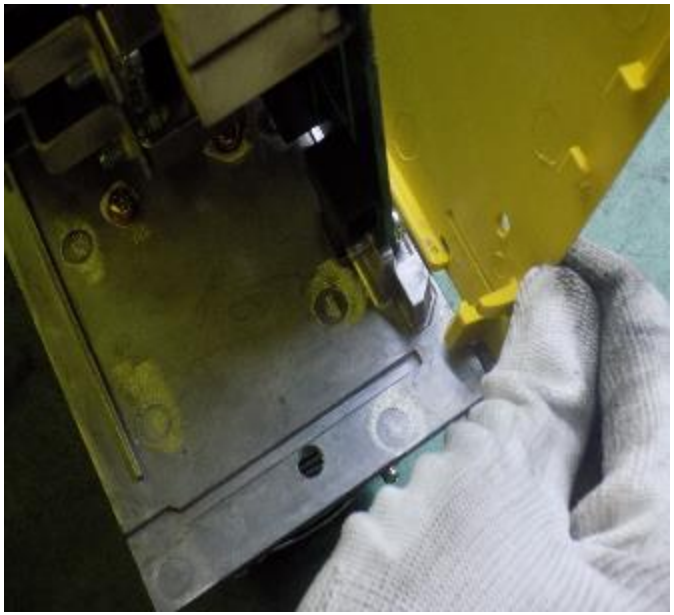
Removal of Plastic Cover RH Side



- In both sides pressed condition, Gently lifted from the left corner of drive
- This allows the side to come out safely from the internal lock within the drive

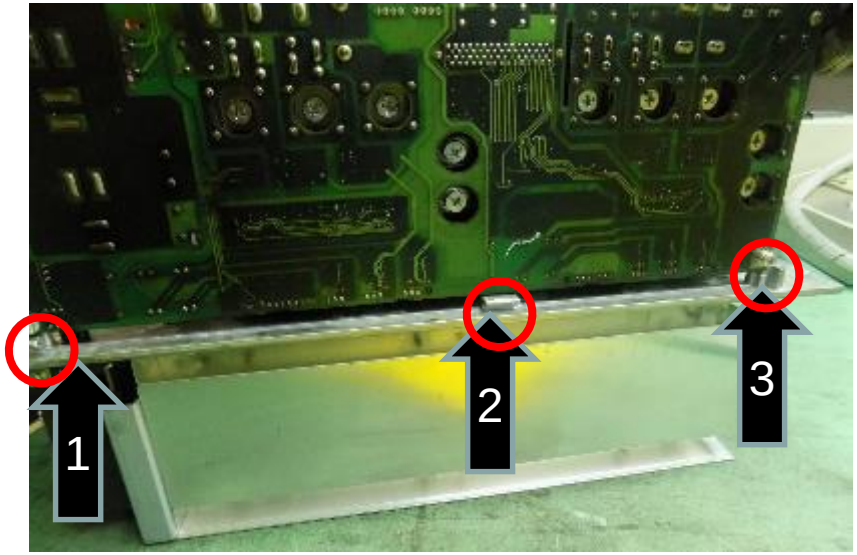
Removal of Plastic Cover RH Side

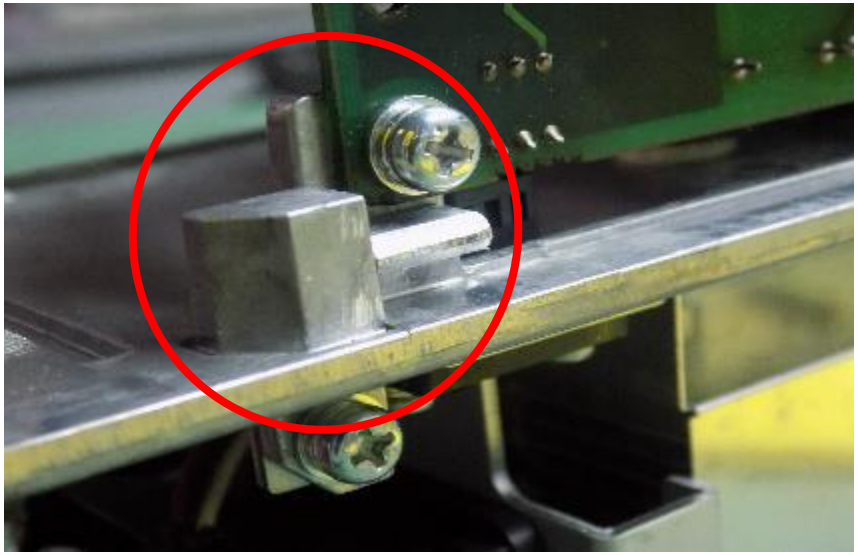




- Then push the right edge of the plastic case to ensure that the lock holding it also get released
- The Side cover – RH case comes out fully from the amplifier base

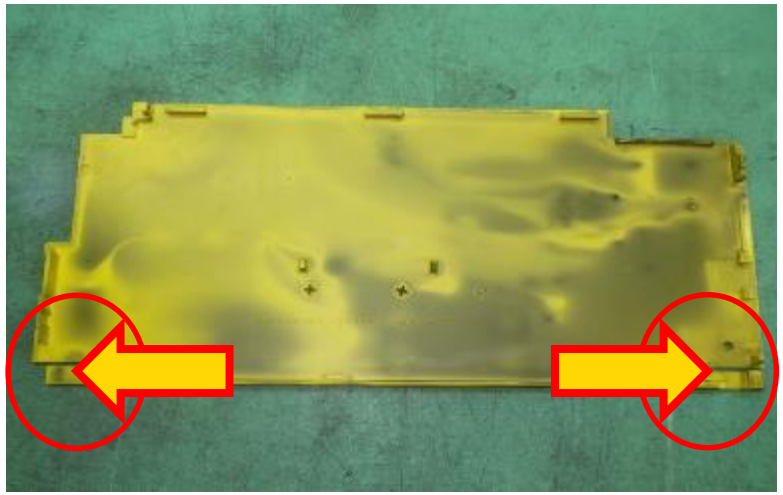
Lock of Plastic Cover RH Side

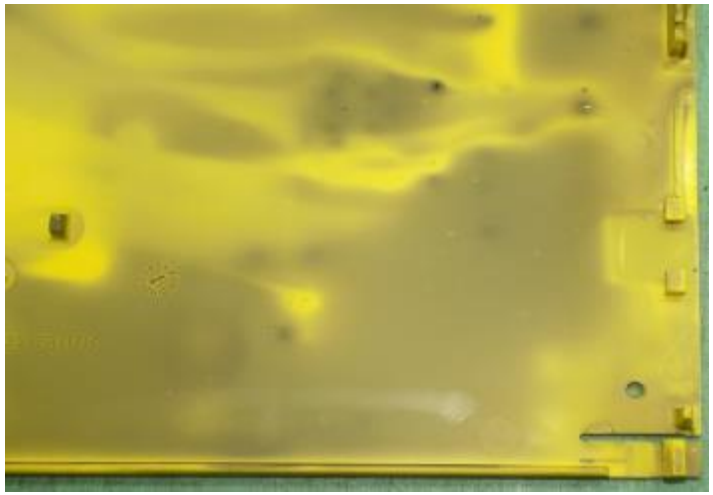




- The above attached picture clearly shows the locks that hold the case from inside
- 1,2&3 are the lock at the middle, left & right edges of the drive base

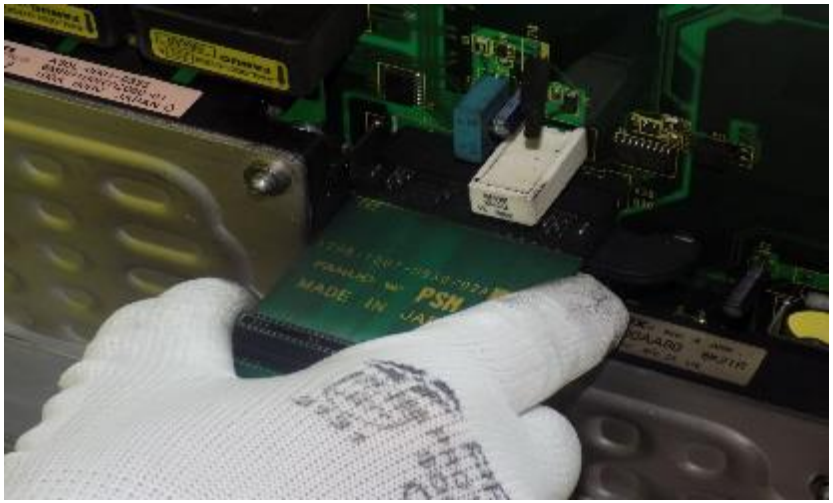
Damaged Plastic Cover RH Side

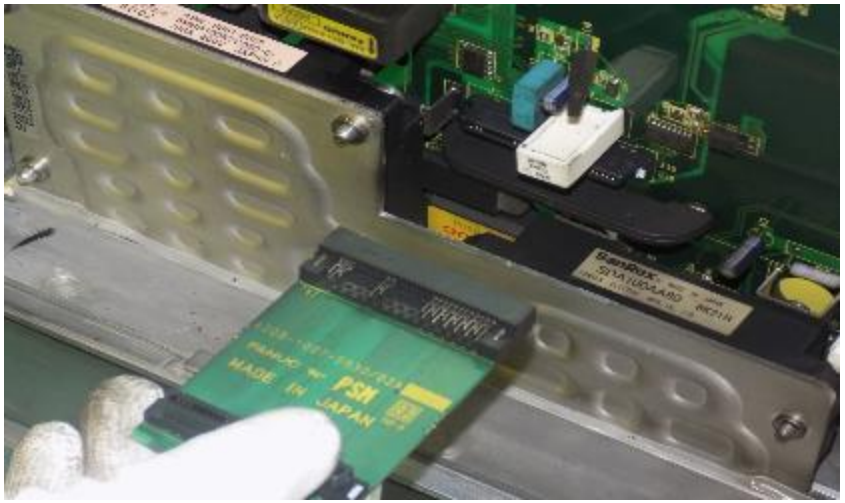




- The above figure shows the internal lock of the case which generally gets damaged or breaks down during its removal
- This phenomena happens if wrong technique is followed to disassemble the side covers

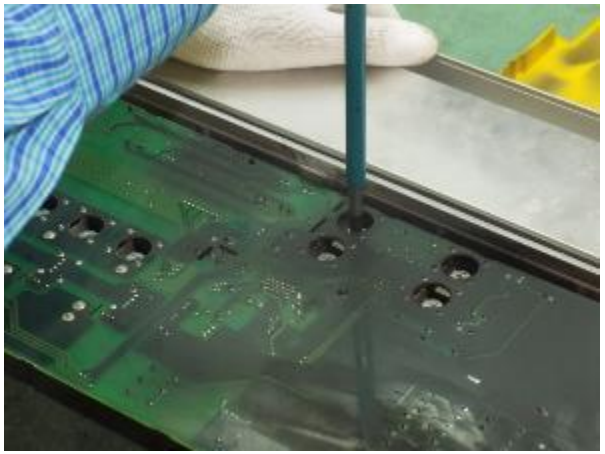
Intermediate PCB dis assembly





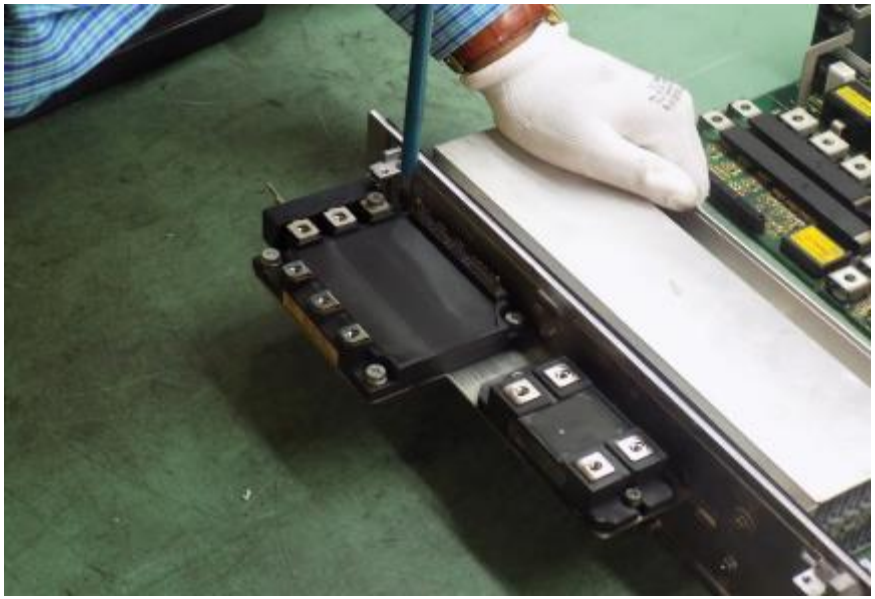
- The next step is to remove the intermediate PCB which is connected to the power & control PCB.
- Catch the pcb from both the edges, pull it outwards and this will allow the pcb to come out of the location on the Power PCB.

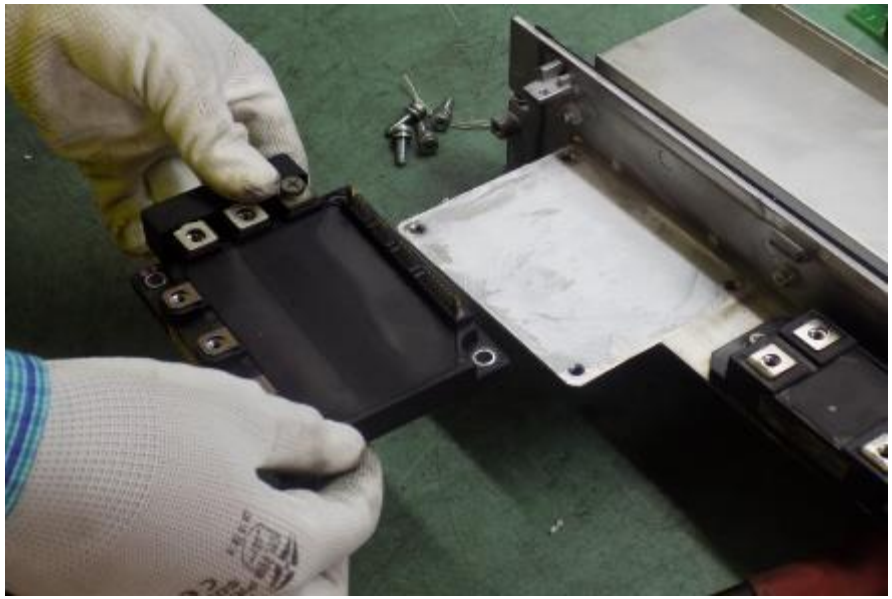
Power PCB dis assembly



- The next step is to remove the Power PCB from the base
- Apply proper pressure and un-screw all 9*M5 star head screws from its mounting using the star screw driver
- Lift the PCB by holding both sides of the Power PCB to dislocate from IPM

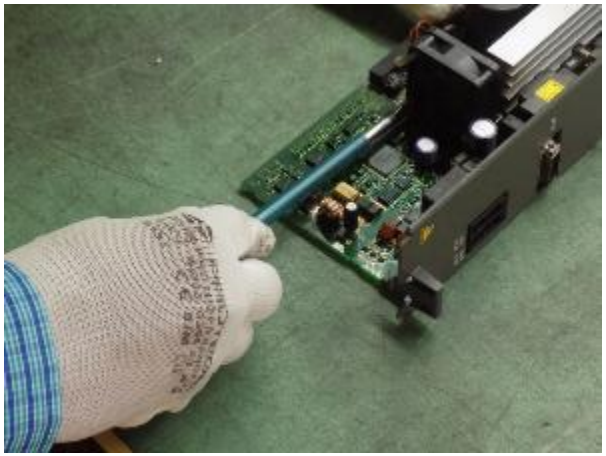
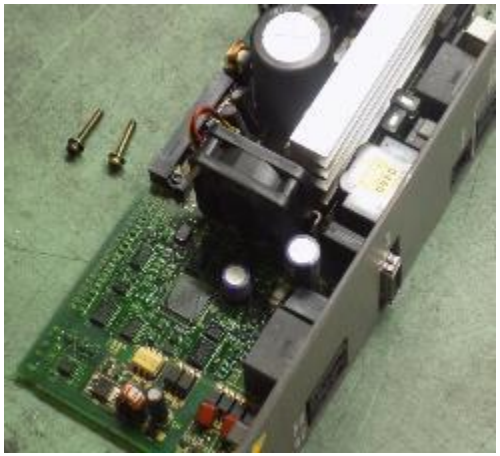
IPM dis assembly





- The final step is to remove the IPM & Thyristor module from the base
- Remove the mounting screw from the side corners and then removing it from its location as shown in the figures

Control PCB Cooling Fan dis assembly

- Final step is to remove cooling fan on the control PCB
- This is generally done by removing the mounting screws from the base as shown in the above figure
- After removing the screws, hold the fan and pull the power connector with cable upwards

THANKING YOU